Investigation of Crushed Concrete Aggregates as a Replacement for Natural Aggregates in AAB Concrete

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1 Introduction

The construction industry is one of the most important industries for a developing country like India as it influences national growth and is a major driver of foreign investment in related industries. Since Portland cement (PC) concrete is a popular building material, demand for it rises in tandem, consequently augmenting its constituents' requirements. It is anticipated that India's planned infrastructural development over the next decade will increase PC production to 800 million tons per annum [1]. This huge amount of PC production results in significant carbon emissions. It is established that 1 ton of PC production results in 0.98 tons of equivalent carbon dioxide emission [2]. The emissions associated with PC production can be reduced by using alternative binders that can partially or completely substitute PC. Alkali-activated binder (AAB) is one such potential alternative that has shown comparable or even superior mechanical and durability properties compared to PC [3].

AAB is produced by the reaction between an aluminosilicate-rich precursor such as fly ash (FA) or ground granulated blast furnace slag (SG) and an alkaline activating solution [3]. The chemistry and the associated reaction mechanisms to form hardened AAB vary with the composition of precursor and activating solution [2, 3]. This complexity in chemistry and the lack of information on long-term properties lead to major hindrances in the widespread practical application of AAB concrete [4]. FA-based AAB concrete requires thermal curing in the range of 60–80 °C to achieve

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A. Sil et al. (eds.), *Recent Trends in Civil Engineering*, Lecture Notes in Civil Engineering 274, https://doi.org/10.1007/978-981-19-4055-2_5

desired mechanical properties, limiting its usage to precast applications [5, 6]. Hence one of the objectives of this study is the development of AAB concrete under ambient-curing conditions.

Another issue associated with the rapid development of global infrastructure is construction and demolition waste (CDW). CDW is dumped into landfills, rendering otherwise usable land unfit for agricultural purposes and impeding sustainable development. Approximately 150 million tons of CDW are generated in India per year [7]. As a result, developing an environmentally sustainable and commercially feasible approach to reuse these CDW is critical. About 90% of the CDW comprises building-derived materials (BDM) such as concrete and bricks. The initiative for recycling CDW began in the early 1980s [8]. The incorporation of recycled coarse aggregates (RCA) in concrete reduced carbon emissions by up to 65% and non-renewable energy consumption by about 58% [9]. According to Serres et al. [10], the energy consumed in RCA concrete and PC concrete productions are 1.39×10^3 MJ and 2.14×10^3 MJ, respectively, and the carbon emissions are 3.35 kg CO_2 /ton and 4.44 kg CO_2 /ton, respectively.

Furthermore, the use of aggregates from CDW has also been investigated with AAB concrete. According to Chindaprasirt and Cao (2015) [11], natural aggregates can be substituted to achieve strengths of more than 70 MPa. The reduction in compressive strength of AAB concrete with recycled aggregates is found to be relatively smaller compared to their conventional concrete counterparts. Analogous to the trend observed in PC concrete, a decline in the strength of FA-based AAB concrete with an increase in percentage substitution of RCA is reported [13]. Nuaklong et al. [14] stated that high calcium FA-based AAB concrete with 100% RCA achieved 78–93% strength compared to limestone-based AAB concrete and suggested optimal molarity of sodium hydroxide (NaOH) as 12. The cracking and susceptibility of recycled aggregate concrete to permeation, diffusion, and absorption of fluids are linked to the cracks and fissures that develop during production. These disadvantages hinder the use of recycled aggregate in reinforced PC concrete at higher percentages (>30%). However, many previous studies reported that AAB concrete has a greater potential for high-volume substitution of RCA than PC concrete without comprising the mechanical properties or durability [11, 15, 16]. The interfacial transition zone in AAB with RCA has higher bond strength compared to AAB with natural aggregates (NA), PC with NA, and PC with RCA, indicating a greater potential to use RCA in AAB [15].

The replacement of NA with RCA in alkali-activated slag mortar significantly reduced the autogenous shrinkage without affecting its strength [12]. The equivalent durability properties with an increase in RCA percentage replacement are attributed to finer binder content in AAB compared to PC. In addition to replacement, researchers focused on improving the properties of BDM with treatments that will improve the properties of concrete made with it. Treatment with supplementary cementitious materials and treatment with carbon dioxide was investigated by Guo et al. [17]. They concluded that treatment with fly ash or other mineral admixtures increases the strength due to reaction between unreacted $Ca(OH)_2$ and fly ash which

forms secondary C-S–H matrix. Due to carbonation, additional $CaCO_3$ is formed which can react with the new mortar and increase the strength.

Most of the research with recycled aggregates was done with conventional concrete. The majority of previous studies have indicated that the optimal replacement percentage for PC concrete is about 30%. Untreated BDM is more eco-friendly than PC, and its potential is not fully explored either in PC or AAB concrete. The majority of previous studies on AAB concrete with RCA used class C FA as a precursor or NaOH with high molarity as an activator. Additionally, chemical treatments have also been used to enhance the quality of recycled aggregates, which raises the cost of the concrete. The present study aims to address the existing gaps in the literature. The principal objectives of this study are: (i) to evaluate the mechanical performance of ambient cured AAB concrete with varying precursor proportion and replacement percentages of virgin RCA as coarse aggregate, (ii) to propose an optimal replacement of RCA for ambient cured AAB concrete, and (iii) correlate the mechanical performance with the observed mineralogical and microstructural changes through X-ray diffraction and scanning electron microscopic analyses. The materials and methods are described below.

2 Materials and Methods

2.1 Materials and Mix Proportions

The crushed virgin BDM used as coarse aggregate in this study is obtained by crushing matured M35 grade concrete to 10 mm nominal size. Crushed granite used as natural coarse aggregate (NA) and river sand conform to ASTM C33/C33M [18]. Class F FA complying with ASTM C618 [19] and SG conforming to ASTM C989/C989M [20] are used as precursors. A combination of NaOH and sodium silicate is used as the activator. Industrial grade NaOH pellets of 99% purity are used. The composition of sodium silicate solution is 55.9% H_2O, 29.4% SiO2, and 14.7% Na2O. Polycarboxylate ether-based superplasticizer (SP) conforming to ASTM C494 [21] type F is used. A total of 20 mixes with four different precursor combinations (FA: SG varied as 100:0, 70:30, 60:40, and 50:50) and five different replacement percentages (0%, 25%, 50%, 75%, and 100%) of NA with BDM are studied. The detailed mix proportions are provided in Table 1.

2.2 Methods

X-ray diffraction (XRD) analysis is performed on powdered BDM samples using a RIGAKU Ultima X-ray Diffractometer to obtain the mineralogical characteristics of the BDM aggregates. The 2θ range is varied from 5° to 90°. The CuKα X-rays are

Table 1 AAB mix designs considered for this study

			FA:SG			
		Mix	100:0	70:30	60:40	50:50
		FA	400	280	240	200
		SG	0	120	160	200
Percentage replacement	0%	NA	1290	1290	1290	1290
		BDM	0	0	0	0
	25%	NA	967.5	967.5	967.5	967.5
		BDM	322.5	322.5	322.5	322.5
	50%	NA	645	645	645	645
		BDM	645	645	645	645
	75%	NA	322.5	322.5	322.5	322.5
		BDM	967.5	967.5	967.5	967.5
	100%	NA	0	0	0	0
		BDM	1290	1290	1290	1290

Unit: kg/m^3; other mix constituents are kept constant as follows, sodium silicate: 129.34 kg/m^3, NaOH: 10.57 kg/m^3, water: 67.65 kg/m^3, fine aggregate: 651 kg/m^3, and SP: 3.142 L/m^3

operated at a voltage of 30 kV and current of 30 mA. The scan rate is maintained at 1°/min with a step size of 0.02°. The results are shown in Fig. 1. For the morphological characterizations, SEM analysis is conducted using an FEI Apreo SEM. The micrographs are captured at an operating voltage of 20 kV and magnification of 2500× (Fig. 2).

For this study, the specific gravity and water absorption of the BDM are determined using the guidelines of IS 2386-3[22]. The results are presented in Table 2 and validated against usage requirements as aggregates in concrete applications.

The activator is allowed to rest for 24 h to alleviate additional heat generated by its exothermic nature. The aggregates are then uniformly mixed with the precursors to obtain a uniform dry mix which is blended with the activator and poured into 100 mm × 100 mm × 100 mm cubes after attaining desired workability. For each mix, three samples are made, and the average compressive strength of these three samples is reported. The specimens are demolded after 24–96 h to allow initial hardening and cured underwater until the age of 7 days from casting. After 7 days, the cubes are placed under ambient conditions till the day of testing. All the tests are performed at the age of 28 days. The compressive strength test is performed using an automatic compression testing machine operated in the load control mode. The load is applied at a rate of 2.33 kN/s.

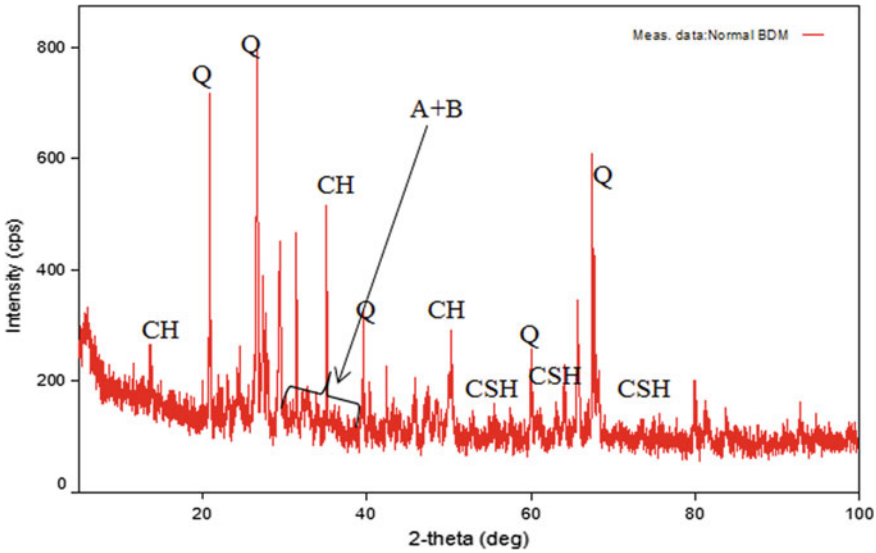

Fig. 1 XRD pattern of the BDM (A = tricalcium silicate, B = di-calcium silicate, CH = calcium hydroxide, CSH = calcium silicate hydrate, Q = Quartz)

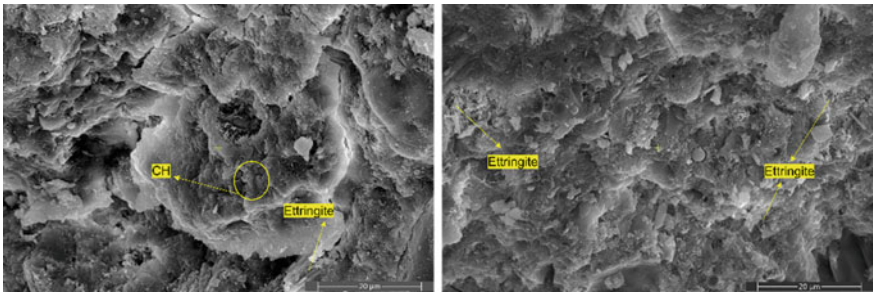

Fig. 2 Scanning electron micrograph of the BDM

Table 2 Water absorption and specific gravity of BDM

Property	Results
Specific gravity	2.54
Water absorption	1%

3 Results and Discussions

Figure 1 presents the XRD pattern of virgin BDM. It reveals the predominance of calcium silicates and quartz in BDM.

The sharp peaks corresponding to 2θ values of 21°, 26°, 39°, 55°, and 66° represent crystalline quartz or silica (Q). The smaller peaks between 2θ values of 30° to 40° represent the tricalcium silicate or alite (A) and dicalcium silicate or belite (B) component of portland cement, whereas the peaks near 2θ values of 16°, 34°, and 49° represent the calcium hydroxide (CH) or portlandite due to hydration of PC. The flatter portions of the diffractogram represented by small humps indicate the presence of an amorphous hydrated cementitious paste component (calcium silicate hydrate, CSH). These results are corroborated with findings from previous research [23].

Figure 2 shows that the morphology of the BDM is composed of a reticular network corresponding to CSH, hexagonal crystals of CH, and slender needle-like crystals of ettringite. These observations are consistent with the findings from hardened pastes of hydrated portland cement [23].

The specific gravity and water absorption results are listed in Table 2. The values obtained for the BDM are within the allowable limits for aggregates used in PC concrete preparation, as specified by IS 2386-3 [22]. The values of specific gravity of aggregates used in PC concrete applications should lie between 2.5 and 3.0. The maximum permissible limit for water absorption of aggregates used in cement concrete applications is 2%.

Figure 3 shows the compressive strength test results of AAB concrete with varying precursor proportions and NA replacement percentages with BDM. With an increase in replacement percentage of NA with BDM from 25 to 50%, the compressive strength of 100:0 and 70:30 decreased by 14% and 11%, respectively. However, with further increase in replacement percentage of NA with BDM to 75% and 100%, the compressive strength of 100:0 increased by 20% and 58%, respectively as the replacement content is increased. Whereas 70:30 exhibited an increase of 10% and 41% in compressive strength, 75% and 100% of NA are replaced with BDM, respectively.

The compressive strength at 100% replacement of NA with BDM is the highest compared to 0% replacement. No significant variation is observed in the compressive

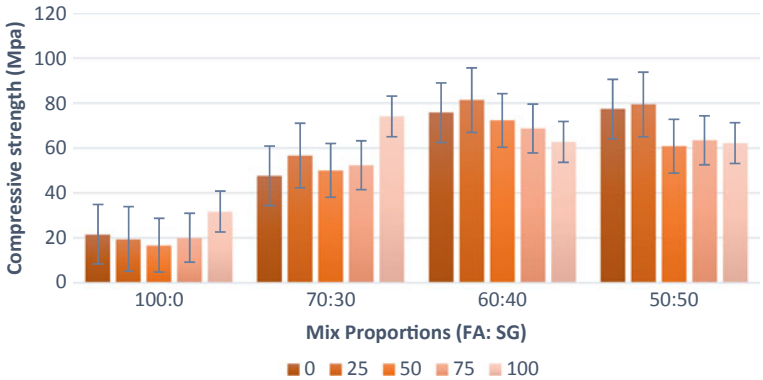

Fig. 3 Compressive strength results for different mix proportions

strength of 60:40 and 50:50 mixes with 25% replacement of NA with BDM. However, a decrease in strength with further replacement of NA with BDM in these mixes is noticed.

According to the compressive strength test data, 100:0 and 70:30 mixes demonstrated an improvement in compressive strength when the NA was entirely replaced with BDM. A comparable pattern is not found in 60:40 and 50:50 mixtures. The authors hypothesize that small spherical FA particles on the surface of crushed aggregates or CSH of virgin BDM reacting with AAB result in a denser matrix. Consequently, the compact matrix leads to an increase in the strength of high fly ash mixes. Further investigation is needed to validate this theory. AAB concrete with FA:SG ratio of 70:30 is proposed as optimum based on the results from the present study. The mix FA:SG of 70:30 exhibits the highest compressive strength at 100% NA replacement with BDM. The selection of AAB concrete with FA:SG of 70:30 is contingent on the primary objective of this study to propose a sustainable concrete mix that can be used for structural applications.

4 Conclusions

In the present study, the mechanical performance of ambient cured AAB concrete with varying precursor proportions, and replacement percentages of NA with BDM is investigated. Based on the findings of this study, the following conclusions are made.

- The primary crystalline phases observed through XRD analysis of BDM are quartz and calcium silicates. SEM analysis of BDM also shows the presence of CSH, CH, and ettringite on the aggregate surface.
- The specific gravity and water absorption show that they are within the limits of IS standard specifications.
- The compressive strength test results indicate that 100:0 and 70:30 blends improve compressive strength by 58% and 41%, respectively, when NA is replaced with BDM at 100%.
- A reduction in compressive strength is observed in 60:40 and 50:50 mixes with increased percentage replacement of NA with BDM.
- Owing to the superior mechanical performance of 70:30 mix at 100% replacement of NA with BDM, it is proposed as the optimum mix.

From the findings of the present study, it is evident that AAB concrete with 100% replacement of NA with BDM can be used for structural applications. Further investigation on the durability of the 70:30 mix and the effect of other BDMs from tiles, bricks, glass, and RAP are recommended.

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